



Introduction to Drones

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Before we begin… let's watch a video



What is Drone?



In Reality, both are drones!!

Drones are flying robot that can be remotely controlled or fly autonomously using software-controlled flight plans in its embedded systems, which work in conjunction with **onboard sensors** and a **Global Positioning Systems (GPS).**

Don’t worry! We will understand everything :)

Section 1



Introduction to Drone

Drone vs UAV vs UAS

- UAV = Unmanned Aerial Vehicle
- UAS = Unmanned Aerial Systems
- Drone = common term
- UAV and UAS are technical terminologies



Unmanned = Uncontrolled?

- No humans inside
- Is controlled by humans remotely
- Follows programmed instructions
- Safety Systems involved



Aircraft vs Drone

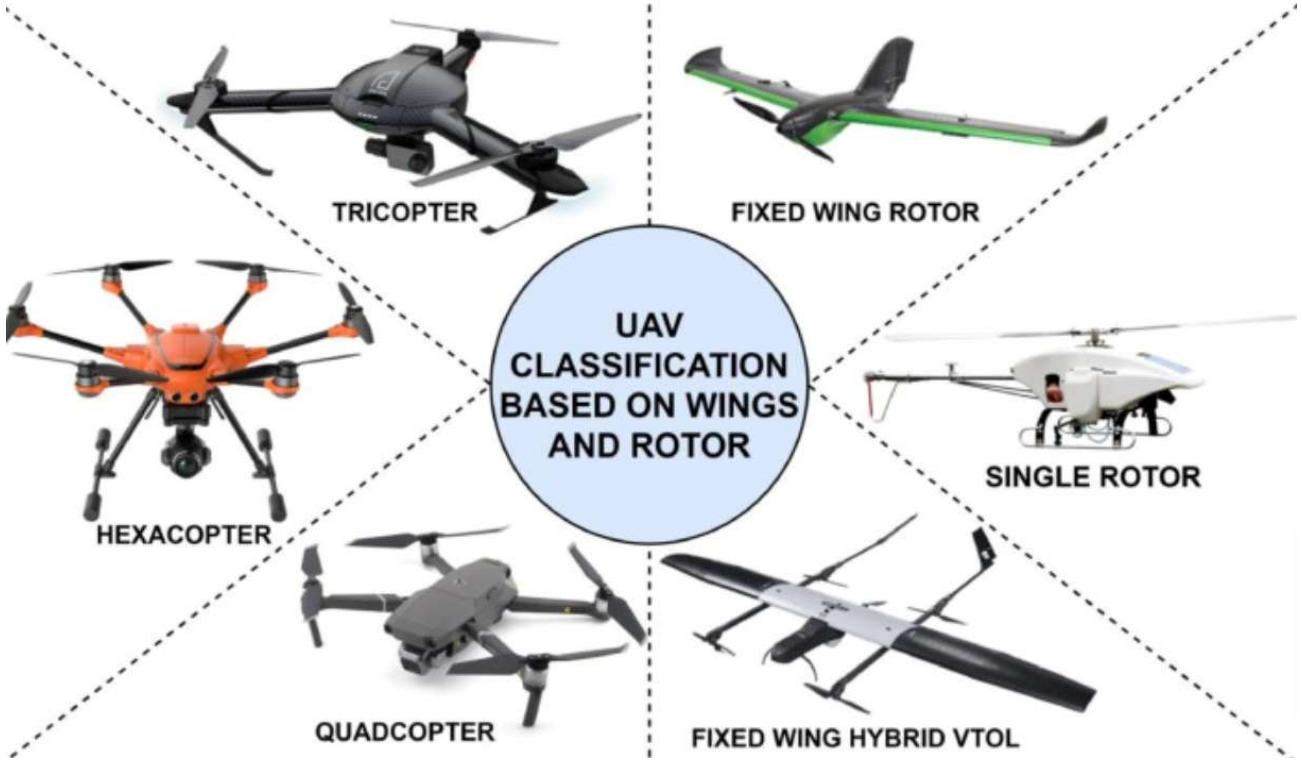


- No pilot onboard
- Can hover
- Vertical takeoff possible
- Smaller, slower



- Pilot onboard
- Cannot hover
- Needs runway
- Larger, faster

Types of Drones



Hybrid Drones



Drones of India



DRDO Abhyas



DRDO Ghatak



DRDO Lakshya



Nagastra



DRDO Bharat



DRDO Imperial Eagle



DRDO Rustom



Heron

and many more…

Application of Drones



Agriculture



Defense

Application of Drones



Construction



Search & Rescue

Application of Drones



Inspection



Inventory Management

Application of Drones

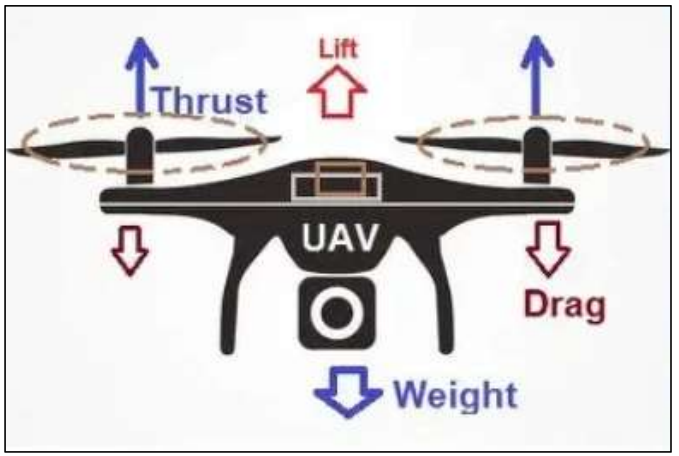


Delivery



Drone Show

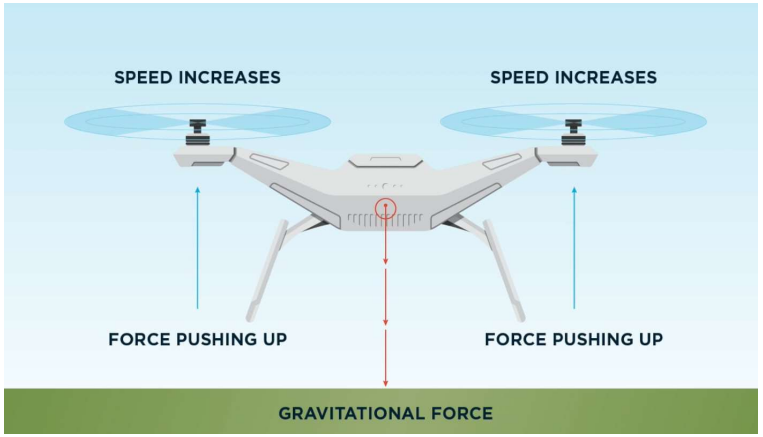
Section 2

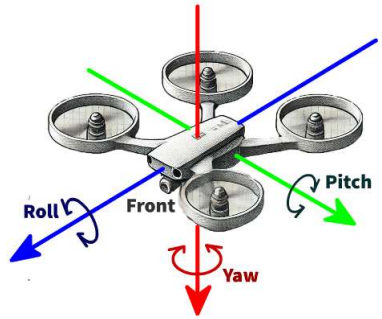


Physics behind Drones

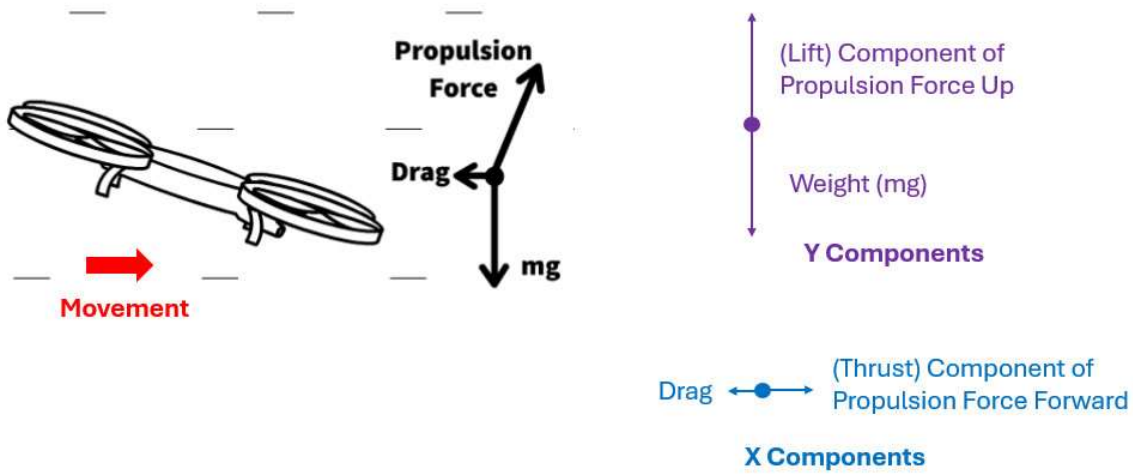
Core Principle of Drone Flight

- **Newton's Third Law of Motion:** For every action, there is an equal and opposite reaction.
- **Why propeller:** Drone propellers push air downwards, they create an upward lift force.
- But at what speed of propeller drone is going to move upward?





Core Principle of Drone Flight



When Lift > Weight

Adjusting the speed of the propellers, drones can finely control their movements and maintain stability during flight.

4 Important Elements of Drone Flying

Propulsion Force/ Thrust

- The propellers of a quadcopter are curved rotate in a way that they push air in one direction.

Drag

- Air resistance that opposes motion

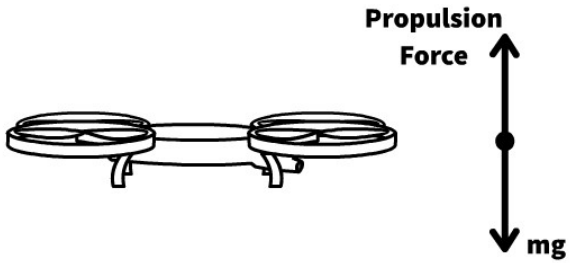
Weight

- The downward pull of the earth on the drones mass equal to the drones weight.
- T = W = mg

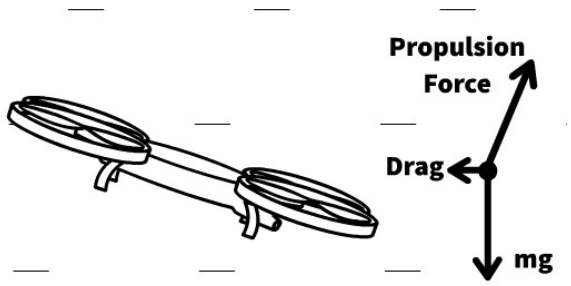
Lift

- The propulsion force of the propellers, when the quadcopter drone is at an angle, can also have a component vertically

4 Elements of Drone Flying



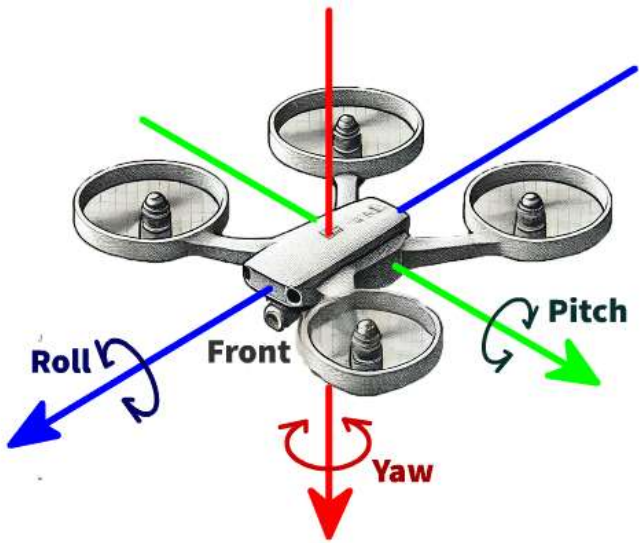
- Drone must be in static equilibrium if it is hovering
- It means there is no net force on drone
- So thrust produce by drone = Weight of drone
- In nutshell, F = W = mg

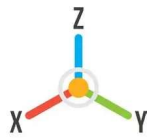


- Drone moving forward
- Propulsion force is tilted forward
- Vertical component balances weight of drone to maintain altitude
- Horizontal component perfectly counteract drag, lead to no net force on drone

Movements that control orientation and direction

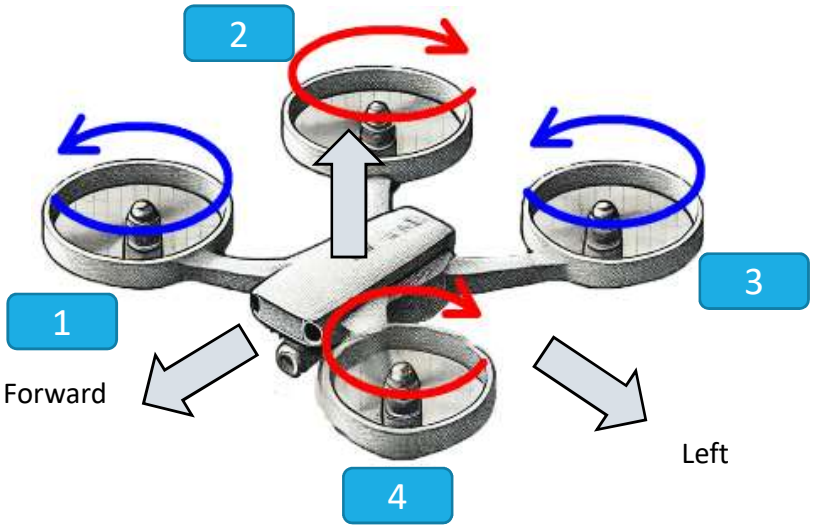
1. **Yaw:** rotation around the vertical axis, allowing the drone to turn left or right
2. **Pitch:** Tilting the drone forward or backward, enabling forward or backward movement
3. **Roll:** tilts the drone sideways, facilitating lateral motion to the left or right





Torque Control

- Adjacent propellers rotate in opposite direction – clockwise and anti-clockwise
- The torques cancel each other out, resulting in a net torque of zero = No drone rotation
- How to move drone in forward direction? **Increase speed of rotor 2 and 3.**
- How to move drone in left direction? **Increase speed of rotor 1 and 2.**
- How to move increase height of drone? **Increase the speed of all rotor at constant speed.**



Section 3



Drone Building

Basic Parts of Drone



Frame

Two main things to note about the frame of a drone.

1. How many arms there are

2. Size of the frame



Motors

Two types of motors: brush and brushless

Look for size and KV rating for selection of motor

Example: Motor size 2213, 935 KV rating



Propellers

Rotary fan connected to motor that helps to produce lift

Comes in 4-digit number: 8045, 1045, 6030 etc

Basic Parts of Drone



Electronic Speed Controller (ESC)

Connects motor, battery and flight controller: to change the speed and direction of drone

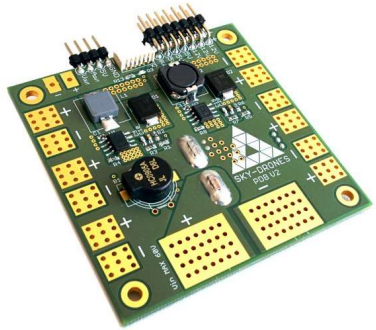
Brushed motors do not require an ESC because they only need a simple DC voltage input.



Battery

Helps to power motor and other electronics

(Lithium Polymer) Lipo battery is most commonly used



Power Distribution Board

Board which allows transferring the power from the battery to ESCs / Motors and generate power supply for the autopilot and other subsystems with different voltage levels.

Basic Parts of Drone



Flight Controller

It integrates sensors, software, and communication modules to control the drone's flight and stability.

Called the "brain" of the drone



Receiver and Transmitter

Transmitter: electronic device that uses radio signals to transmit commands wirelessly

Receiver: connected to drone which receives signals from transmitter



Telemetry

Continuous stream of real-time flight, health, and sensor data sent between a UAV and its ground control system.

Tools required for drones

1. **Soldering Iron and solder:** Soldering iron for assembling some of drone parts
2. **Multi-meter:** Help in diagnosing any electrical problems
3. **Double sided sticky tape:** Fixing many parts to frame simply with double sided sticky tape.
4. **Velcro:** Helpful for fixing lipo battery to main frame
5. **Zip Ties:** Easier to fix to the frame with simple zip ties, like ESCs
6. **Heat shrink or electrical tape:** It provide insulator around your metal bullet connectors



Velcro



Zip Ties



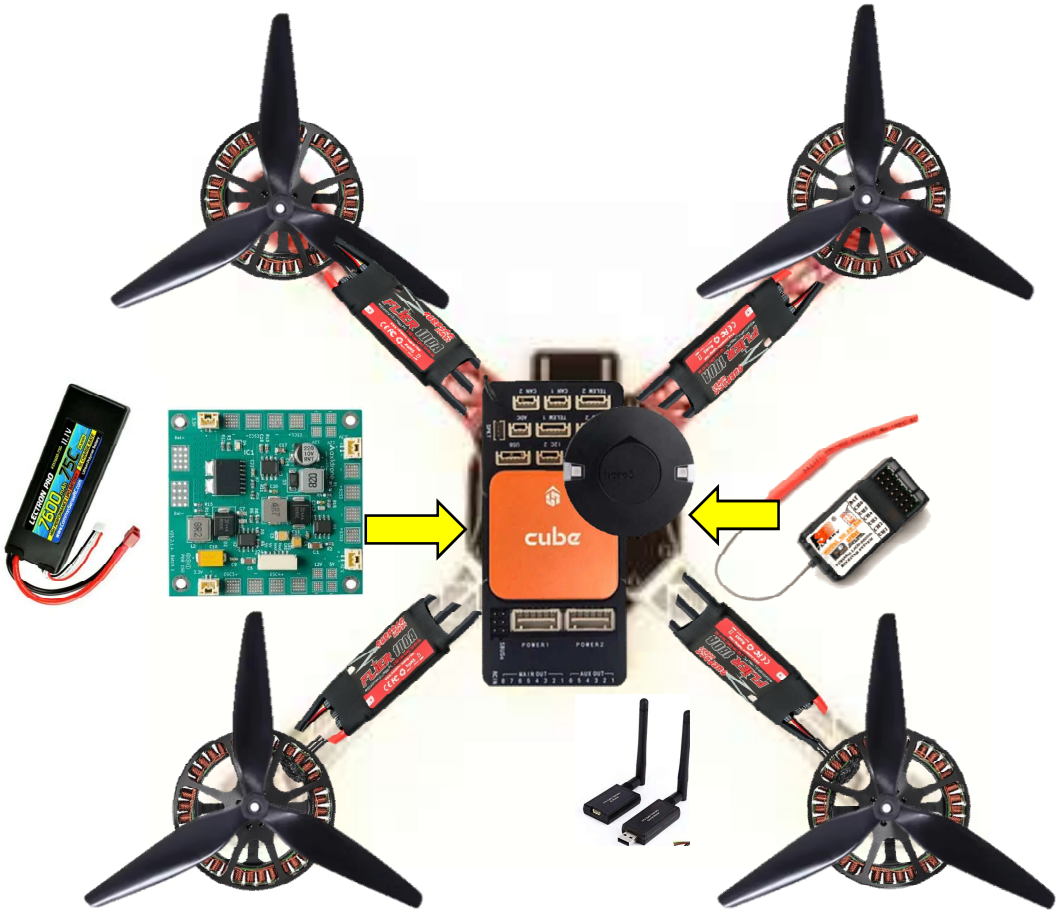
electrical tape



Multi-meter

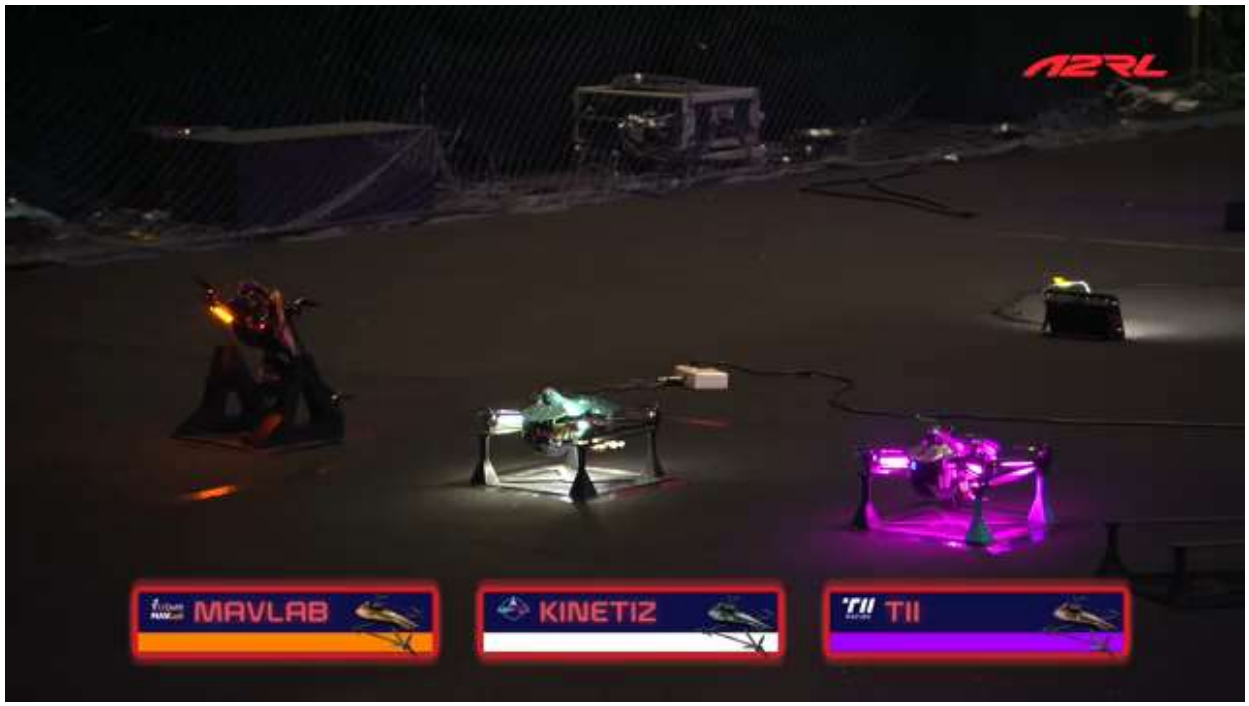


Soldering Iron and solder


cube



Lot of Theory, Let’s go to Stadium!!



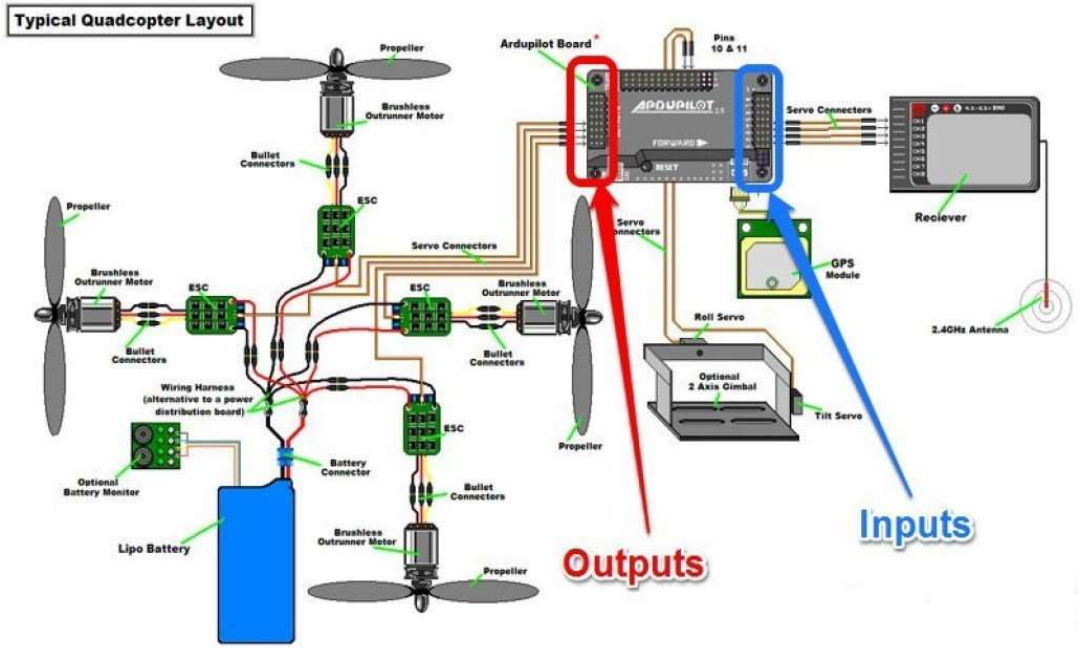
Section 4



How Drones Fly

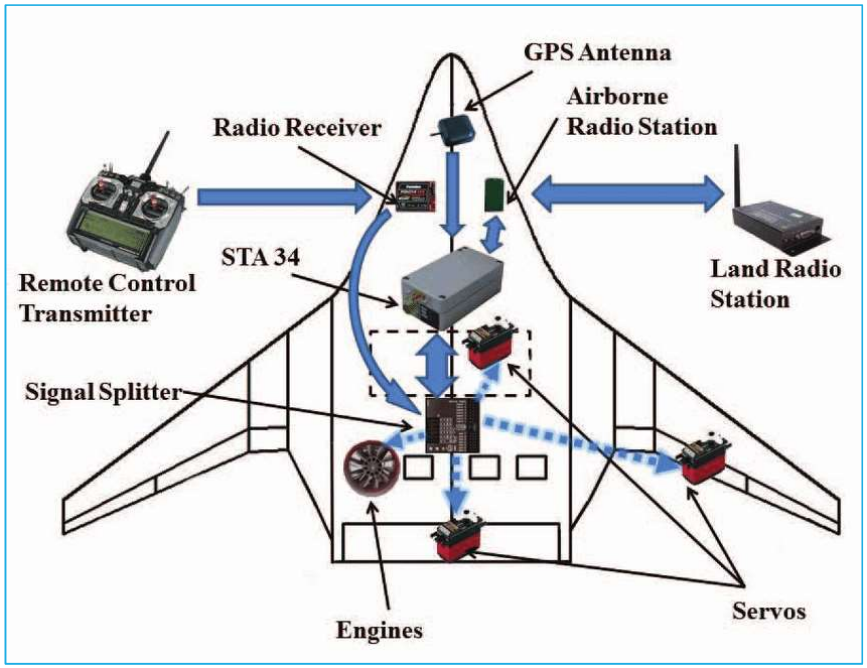
Flight Control System

- Brain of Drone
- Integrates sensors, processors, motors
- Ensures drone remains stable and responsive

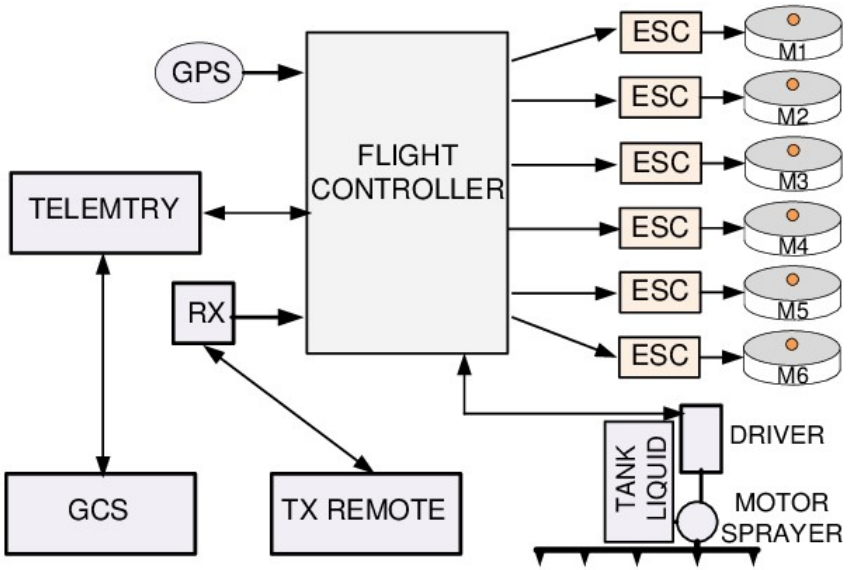


Flight Control System





Input and Output to Flight Controller



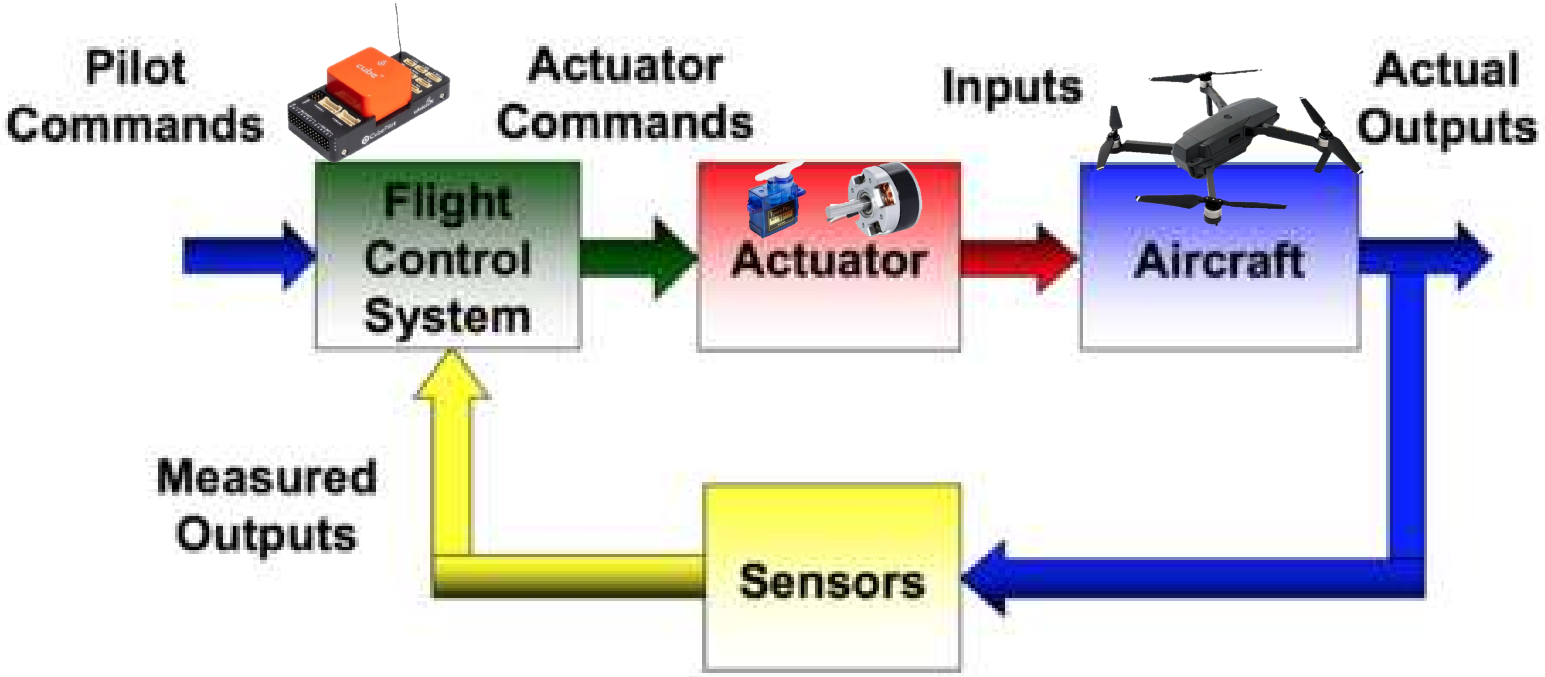
Input

- Angular Velocity (Gyroscope)
- Acceleration (Accelerometer)
- Orientation (GPS)
- Position (GPS)

Output

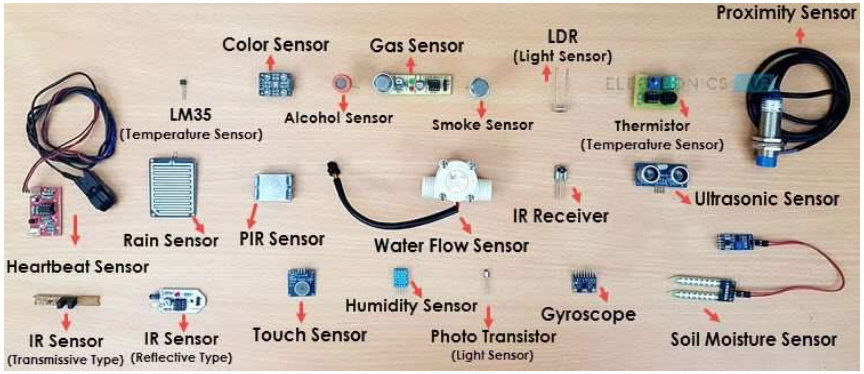
- Thrust and Torque (Converted to Motor speeds)

Flight Control System Block Diagram



Sensors

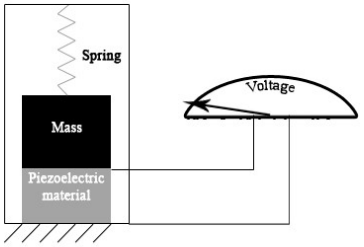
- Device that converts signals from one energy domain to electrical domain.
- Can detect and response to changes in the environment.
- Changes can be in form of light, temperature, motion, moisture or any other physical property.
- In Drones it is used to measure the orientation and position.

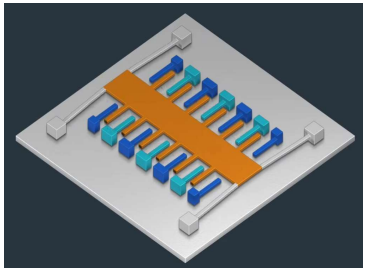


Accelerometer

- Device that measures acceleration
- Can measure acceleration in 1,2 or 3 axis
- Consists of capacitive plates
- Plates move -> capacitance changes -> voltage change



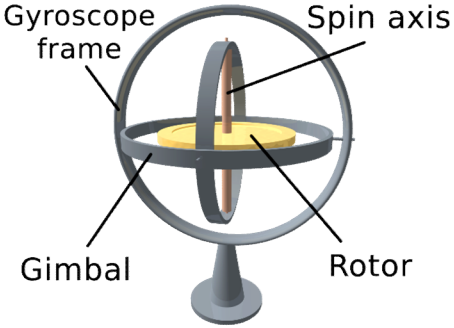


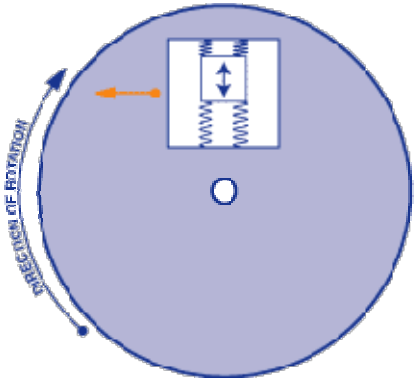


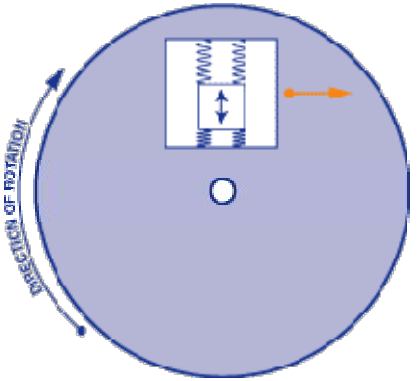
Gyroscope

- Device that measures rotational motion
- Used to determine orientation
- Gyro rotated -> Angular velocity changes -> Small Mass shifts -> Movement converted to low-current electric signal -> read by micro-controller





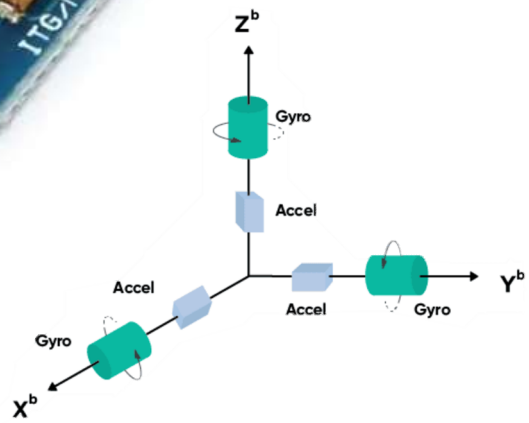




Inertial Measurement Unit (IMU)

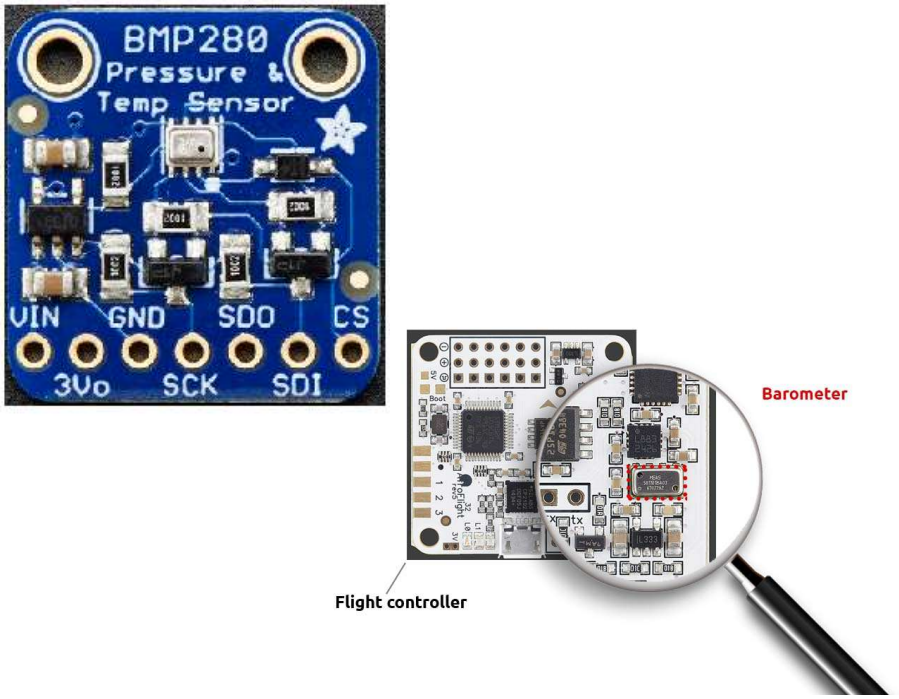
- Combination of Accelerometer and Gyroscope
- Detect Linear Acceleration and Angular Rate in about X,Y and Z axes
- Small size and more cost effective





Barometer

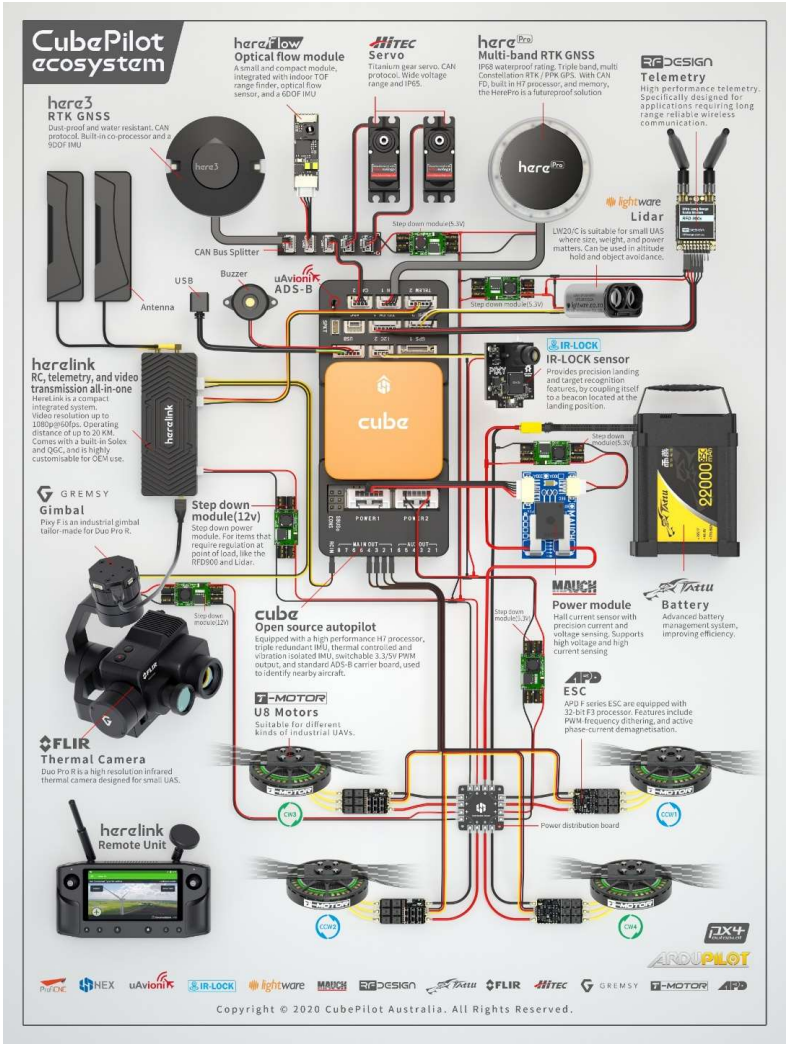
- Gives the height/altitude of the aircraft
- Measures change in atmospheric pressure
- Principle – As altitude increases, air pressure decreases and vice versa



GPS

- Satellite system that allows to determine precise geographic location
- Helps in accurate navigation and autonomous flight
- With GPS, drone fly with complete autonomy and follow pre-defined routes



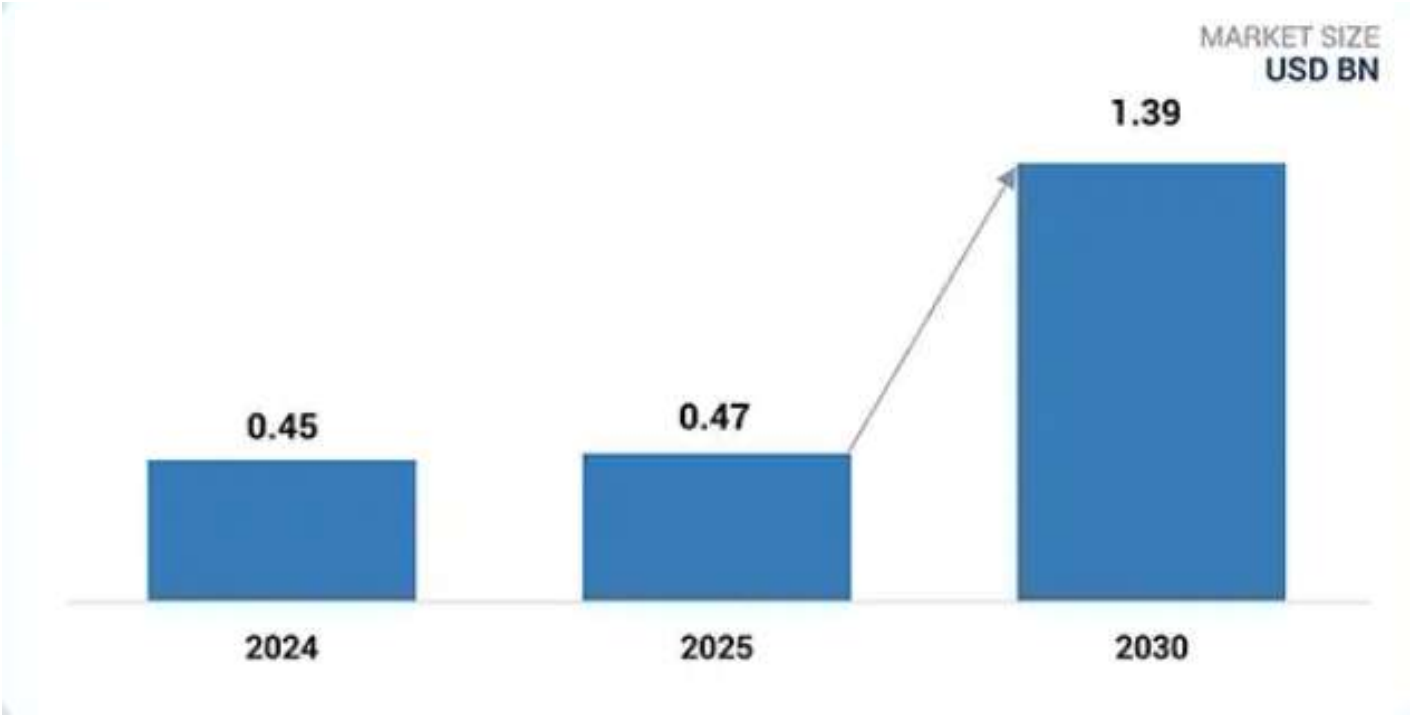

CubePilot ecosystem
here3 RTK GNSS
Dust-proof and water resistant. CAN protocol. Built-in co-processor and a 9DOF IMU
here flow Optical flow module
A small and compact module, integrated with indoor TOF range finder, optical flow sensor, and a 6DOF IMU
Hitec Servo
Titanium gear servo. CAN protocol. Wide voltage range and IP65.
here Pro Multi-band RTK GNSS
IP68 waterproof rating. Triple band, multi Constellation RTK / PPK GPS. With CAN FD, built in H7 processor, and memory, the HerePro is a futureproof solution
RFDESIGN Telemetry
High performance telemetry. Specifically designed for applications requiring long range reliable wireless communication.
lightware Lidar
LW20/C is suitable for small UAS where size, weight, and power matters. Can be used in altitude hold and object avoidance.
CAN Bus Splitter
Step down module(5.3V)
USB
Buzzer
uAvionix ADS-B
Antenna
herelink RC, telemetry, and video transmission all-in-one
HereLink is a compact integrated system. Video resolution up to 1080p/960fps. Operating distance of up to 20 KM. Comes with a built-in Solex and QGC, and is highly customisable for OEM use.
IR-LOCK IR-LOCK sensor
Provides precision landing and target recognition features, by coupling itself to a beacon located at the landing position.
cube
GREMSY Gimbal
Pixy F is an industrial gimbal tailor-made for Duo Pro R.
Step down module(12v)
Step down power module. For items that require regulation at point of load, like the RFD900 and Lidar.
MAUCH Power module
Hall current sensor with precision current and voltage sensing. Supports high voltage and high current sensing
Tattu Battery
Advanced battery management system, improving efficiency.
cube Open source autopilot
Equipped with a high performance H7 processor, triple redundant IMU, thermal controlled and vibration isolated IMU, switchable 3.3/5V PWM output, and standard ADS-B carrier board, used to identify nearby aircraft.
T-MOTOR U8 Motors
Suitable for different kinds of industrial UAVs.
APD ESC
APD F series ESC are equipped with 32-bit F3 processor. Features include PWM frequency dithering, and active phase-current demagnetisation.
FLIR Thermal Camera
Duo Pro R is a high resolution infrared thermal camera designed for small UAS.
herelink Remote Unit
Power distribution board
CW1
CCW1
CW2
CCW2
CW3
CW4
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Section 5



Indian Drone Market

Indian Drone Market



Drone Companies in India (As of 2023)



Drone Companies in India



Section 6



Conclusion

What we learned today?

- What is drone and what are the various sectors it is used
- How the drones fly and hover and maneuver in the air
- Forces and Moments involved in the movement of drone
- Components that are used to build a small drone and its assembly
- Flight Control Systems and Sensors put in the drone for autonomy
- Indian Drone Companies and Drone Market

Get Ready for Tomorrow!!



We now know how to fly but we don't want our drone to get lost!!

Thank You ☺